# Robert Chen

Centreville, VA | (240) 898-8894 | robert@robertwchen.com | robertwchen.com | linkedin.com/in/robertwchen | Active Public Trust

## Education

**University of Virginia** | GPA: 3.6/4.0 | Charlottesville, VA
*B.S. in Computer Engineering and Electrical Engineering (Double Major)* | May 2026

## Experience

**Software Engineer (Analyst)** — Arlington, VA
*Accenture Federal Services* — *Aug 2026 – Present*

- Designed the architecture for a benefits processing microservice, analyzing about 1,800 business rules to determine when rules could run once or needed reevaluation.
- Implemented a Java/Spring Boot REST service that invokes a rules engine to evaluate rules from an AI pipeline. Matched legacy results in **all 28 API integration tests** for a benefits workflow and added Excel-based regression tests to the Maven build.
- Developed a debugger to trace rule execution through the new API.
- Improved notification error reporting and standardized phone formats across 3 Spring Boot services, adding unit tests.

**DevOps Cloud Automation Engineer Intern** — Fairfax, VA
*Verint* — *May 2025 – Nov 2025*

- Deployed a secure AWS Transfer Family web client on ECS Fargate using CloudFormation, Cognito, and Angular.
- Won Best Intern Project out of 6 teams for a banking assistant prototype with voice biometrics and compliance guardrails. The team's business case projected about $10M in annual savings.
- Cut CI/CD checks from 3–5 to **2–3 minutes** for about 30 pull requests per week using GitHub Actions and Harness.
- Improved PR-check precision by **12%** by curating pull-request data and tuning an Azure AI Foundry code review model.
- Migrated Terraform workspaces to OpenTofu, standardizing remote state and reusable modules for EC2, S3, and IAM.

**Research Assistant (ML & Computer Vision)** — Charlottesville, VA
*Visual Intelligence Lab, University of Virginia* — *Sep 2025 – May 2026*

- Parsed tennis broadcasts into JSON match data with Python, OpenCV, Tesseract OCR, smoothing, and finite-state logic.

**Software Engineering Intern** — Centreville, VA
*S&J Chen, LLC* — *May 2024 – Aug 2024*

- Reduced manual rent follow-ups by about **60%** across 2 shopping centers by shipping a full-stack tenant payment portal with Next.js, TypeScript, Supabase, and Vercel.
- Cut monthly reporting from about 3 hours to **30 minutes** by automating Stripe webhook reconciliation and CSV exports.

## Projects

**Badminton Swing Analyzer** | *Embedded C, Flutter, BLE, IMU, SQLite*

- Built Embedded C firmware and a Flutter app using 100 Hz IMU and audio data, achieving about **95%** hit-vs-shadow-swing accuracy in on-court tests with FFT and physics-based features.

**Photo Curator App** | *React Native, TypeScript, TensorFlow.js, NestJS, AWS*

- Built a React Native photo curator with swipe review, blur detection, and facial recognition, backed by a NestJS API with secure sync and **sub-200 ms** similarity search (JWT, pgvector, S3).

**Outreach Ops Agent** | *n8n, OpenAI API, Google Sheets, Clay, Gmail API, Docker*

- Automated outreach research and drafting with retry-safe queues and human approval before sending, cutting average hands-on prep from **about 15 minutes to 1 minute** per posting in a timed trial.

## Technical Skills

**Programming Languages:** Java, Python, TypeScript, JavaScript, SQL, C, C++
**Frameworks & Tools:** Spring Boot, REST APIs, OpenAPI, Maven, JUnit, Mockito, Kafka, React, Next.js, Angular, Node.js, Git
**ML & Data:** PostgreSQL, pgvector, Supabase, OpenCV, Tesseract OCR, TensorFlow.js, Azure AI Foundry, NumPy, Pandas
**Cloud & DevOps:** AWS (ECS, S3, Cognito), Docker, Kubernetes, GitHub Actions, Harness, Terraform, OpenTofu, Linux

## Leadership & Activities

**Leadership:** Yo-Yo Club, Founder & President (about 50 members); Taiwanese Student Association (TSA), Sports Chair
**Athletics:** Club Pickleball (Collegiate Nationals); Club Tennis (Competitive Travel Team); Club Figure Skating (2× 1st Place)